加：

```python
class Complex:
    def __init__(self, re, im):
        self.real = re
        self.imag = im

    def __add__(self, other):
        return Complex(self.real + other.real, self.imag + other.imag)
        return result

    def __repr__(self):
        return 'real = %0.2f, imag = %0.2f' % (self.real, self.imag)

c1 = Complex(3, -4)
c2 = Complex(-3, 4)

c3 = c1 + c2

print(c1)
print(c2)
print(c3)
```

问题 输出 调试控制台 终端

```
PS E:\Advanced programming> & E:/miniconda/python.exe "e:/Advanced programming/Complex.py"
real = 3.00, imag = -4.00
real = -3.00, imag = 4.00
real = 0.00, imag = 0.00
```

减：

```python
class Complex:
    def __init__(self, re, im):
        self.real = re
        self.imag = im

    def __sub__(self, other):
        # (a+bi) - (c+di) = (a-c) + (b-d)i
        return Complex(self.real - other.real, self.imag - other.imag)

    def __repr__(self):
        return 'real = %0.2f, imag = %0.2f' % (self.real, self.imag)

c1 = Complex(3, -4)
c2 = Complex(-3, 4)

c3 = c1 - c2

print(c1)
print(c2)
print(c3)
```

问题 输出 调试控制台 终端

```
PS E:\Advanced programming> & E:/miniconda/python.exe "e:/Advanced programming/c
ramming/complex_sub.py"
real = 3.00, imag = -4.00
real = -3.00, imag = 4.00
real = 6.00, imag = -8.00
```

乘：

```python
class Complex:
    def __init__(self, re, im):
        self.real = re
        self.imag = im

    def __mul__(self, other):
        # (a+bi)(c+di) = (ac-bd) + (ad+bc)i
        re = self.real * other.real - self.imag * other.imag
        im = self.real * other.imag + self.imag * other.real
        return Complex(re, im)

    def __repr__(self):
        return 'real = %0.2f, imag = %0.2f' % (self.real, self.imag)

c1 = Complex(3, -4)
c2 = Complex(-3, 4)

c3 = c1 * c2

print(c1)
print(c2)
print(c3)
```

问题 输出 调试控制台 终端

```
real = 6.00, imag = -8.00
PS E:\Advanced programming> & E:/miniconda/python.exe "e:/Advanced programming/c
omplex_mul.py"
real = 3.00, imag = -4.00
real = -3.00, imag = 4.00
real = 7.00, imag = 24.00
```

除：

```python
class Complex:
    def __init__(self, re, im):
        self.real = re
        self.imag = im

    def __truediv__(self, other):
        # (a+bi)/(c+di) = (ac+bd)/(c^2+d^2) + (bc-ad)/(c^2+d^2)i
        denominator = other.real ** 2 + other.imag ** 2
        if denominator == 0:
            raise ZeroDivisionError('cannot divide by zero complex number')
        re = (self.real * other.real + self.imag * other.imag) / denominator
        im = (self.imag * other.real - self.real * other.imag) / denominator
        return Complex(re, im)

    def __repr__(self):
        return 'real = %0.2f, imag = %0.2f' % (self.real, self.imag)

c1 = Complex(3, -4)
c2 = Complex(-3, 4)

c3 = c1 / c2

print(c1)
print(c2)
print(c3)
```

问题 输出 调试控制台 终端

```
real = -3.00, imag = 4.00
real = 7.00, imag = 24.00
PS E:\Advanced programming> & E:/miniconda/python.exe "e:/Advanced programming/complex_div.py"
real = 3.00, imag = -4.00
real = -3.00, imag = 4.00
real = -1.00, imag = 0.00
```